

Analyzing Incident Management Practices in Global IT Service Management

An Expert perspective on Challenges, Opportunities, and ITIL4 Alignment

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Abstract—In today’s complex, global IT environment, effective Incident Management (IM) is crucial for maintaining service continuity and client trust. While theoretical frameworks like ITIL 4 provide guidance, real-world application often faces distinct operational challenges. The objective of this study is to bridge the gap between theoretical frameworks and real-world execution by analyzing the practical application of IM at scale. The methodology combines a literature review with a semi structured expert interview conducted with a Program Transformation Director possessing over 25 years of experience in the private sector. The empirical data was analyzed using a SWOT framework (Strengths, Weaknesses, Opportunities, and Threats) to categorize internal and external factors. The results indicate that while organizations demonstrate strengths in SLA compliance and process consistency, they face weaknesses regarding internal metric misalignment, such as premature ticket closure and communication gaps with clients. Furthermore, while opportunities were identified in the adoption of AI (Artificial Intelligence) and L1.5 support models to enhance LFR (First Contact Resolution), external threats regarding commercial pricing pressures and staff attrition pose significant risks to service stability. The study concludes that aligning internal metrics with actual value creation and investing in staff stability are essential for adhering to ITIL 4 principles in a global context.

Keywords—Incident Management; IT Service Management; ITIL4; IT Outsourcing; Escalation; Communication; Best Practices (key words)

I. INTRODUCTION

The modern digital economy relies heavily on the continuous availability of IT services. In this environment, IT operations grapple with the constant challenge of minimizing service disruptions while maintaining complex, global systems. Incident Management (IM)

serves as the most immediate and critical practice within this domain, defined by the swift action of minimizing the negative impact of an incident and restoring normal service operation as quickly as possible. This thesis explores the operational reality of this practice and its alignment with established theoretical standards.

II. BACKGROUND

The field of IT operations grapples with the challenge of minimizing service disruptions while maintaining complex systems. The need for a standardized approach led to the formalization of the IT Service Management (ITSM), which provides a strategic framework for managing the design, delivery, and improvement of IT services (AXELOS, 2019). ITSM is not merely a set of tools but a comprehensive approach to ensuring that IT delivers demonstrable value to the business and its customers (Marrone & Kolbe, 2011)

A. Problem Formulation

While frameworks like ITIL 4 provide comprehensive guidelines, such as “Focus on value” and “Collaborate and promote visibility” (AXELOS, 2019), there is often a disconnect between these theoretical principles and the operational reality of global IT service providers. Academic literature outlines the theoretical success models for ITSM implementation (Lau et al., 2017). However, fewer studies utilize expert perspectives to analyze why operational metrics and human factors often cause deviations from these standards in a high-volume, global context.

The problem is that organizational and human elements of the ITSM adoption are often more challenging than the theoretical framework itself (Marrone & Koble, 2011).

There is a need to bridge the gap between theoretical frameworks and real-world execution by analyzing the practical application of IM at scale.

B. Objective

The objective of this study is to bridge the gap between theoretical frameworks and real-world execution by analyzing the practical application of IM at scale. By integrating theoretical perspectives with expert insights, this study aims to identify where real-world processes align with or diverge from established ITIL guidelines.

C. Research Questions

To achieve this objective, the study focuses on the following research questions:

- RQ1: What are the key internal and external factors (Strengths, Weaknesses, Opportunities, and Threats) that characterize effective Incident Management practice in large organizations?
- RQ2: How does the practical application of Incident Management align with, or diverge from, the core principles of the ITIL 4 framework?

D. Delimitation

This study is delimited to the perspective of a single global IT service provider context, utilizing a qualitative expert interview to gain insight into high-volume service environments. The research focuses specifically on Incident Management practice and does not cover other ITSM practices such as Change Management. Furthermore, the analysis is restricted to the comparison between the ITIL 4 framework and the specific operational realities identified in the case study.

III. THEORETICAL FRAMEWORK

This chapter establishes the academic foundation for the study. It explores the definitions of Incident Management within the ITIL framework, outlines the tiered support structure, and discusses the specific challenges of global IT service delivery. Finally, it presents the SWOT framework used to analyze

A. IT Service Management and ITIL 4

The need for a standardized approach to IT operations led to the formalization of IT Service Management (ITSM). ITSM is defined as a strategic framework for managing the design, delivery, and improvement of IT services. It is not merely a set of tools but a comprehensive approach to ensuring that IT delivers demonstrable value to the business and its customers (Marrone & Kolbe, 2011)

Central to ITIL 4 is the Service Value System (SVS).

Agutter (2020) explains that the SVS demonstrates how the various components and activities of the organization work together to facilitate value and creation through IT-enabled services. This system includes the Service Value Chain, practices, governance, and guiding principles.

ITIL 4 outlines seven guiding principles that can be adopted at all organizational levels. According to Agutter (2020), these principles are universally applicable and essential for decision-making. This study focuses on three:

- **Focus on value:** Every action taken during an incident should be traceable back to restoring service and delivering value to the customer (AXELOT, 2019).
- **Collaborate and promote visibility:** Effective incident resolution requires clear, transparent communication among all involved parties, both internal teams and external stakeholders (AXELOT, 2019).
- **Optimize and automate:** Organizations should maximize the value of human resources by automating standard tasks and utilizing technology to streamline workflows (AXELOT, 2019).

B. Incident Management (IM)

Incident Management (IM) is the most immediate and critical practice within ITSM. According to AXELOS (2019), an incident is an unplanned interruption to a service or a reduction in the quality of a service. The primary goal of IM to minimize the negative impact of an incident and restore normal service operation as quickly as possible.

To manage the volume and complexity of issues, IT organizations typically organize support into a tiered hierarchy. This ensures a structured approach to problem resolution where issues are escalated based on difficulty (AXELOS, 2019).

- **Level 1 (L1) Support:** Often referred to as the Service Desk, this is the initial point of contact for end-users. The focus is on quick resolution (First Contact Resolution) or accurate categorization.
- **Level 2 (L2) Support:** If an incident cannot be resolved at L1, it is escalated to L2. This level possesses more in-depth technical knowledge and access rights.
- **Level 3 (L3) Support:** This level typically involves subject matter experts, developers, or vendors capable of resolving complex systemic issues.

C. Challenges in Global IT Environments

Implementing these frameworks in a global environment introduces specific challenges. The academic literature identifies that minimizing service disruptions while maintaining complex systems is a primary struggle for IT operations.

- **Communication vs. Technical Resolution:** A key tension in IM is balancing the technical fix with stakeholder communication. Research suggests that the organizational and human elements of ITSM adoption—such as communication gaps—are often more challenging than the theoretical framework itself (Marrone & Kolbe, 2011).
- **Global Distribution:** Large organizations often utilize “Follow-the-Sun” models, which require seamless handovers between regional teams. This introduces risks regarding consistency and knowledge transfer that pure process theory may not fully mitigate (Marrone & Kolbe, 2011).

D. Analytical Framework: SWOT Analysis

To analyze the gap between theoretical principles and practical execution, this study utilizes a conceptual framework. Oates (2006) defines a conceptual framework as a structure that makes explicit the researcher’s way of thinking about the topic. In this study, the SWOT framework (Strengths, Weaknesses, Opportunities, and Threats) is employed to categorize the qualitative data.

- Strengths and Weaknesses address internal factors, such as process consistency or metric misalignment.
- Opportunities and Threats address external factors, such as technology adoption, like Artificial Intelligence or commercial risks like staff attrition.

This framework allows for a structured assessment of the organization’s resilience and its alignment with ITIL 4 principles.

IV. METHODOLOGY

This study employs a qualitative, dual-methodology approach combining literature review with an expert interview to gather comprehensive insights into Incident Management practices. This mixed approach ensures that the analysis is grounded in both established theoretical frameworks and contemporary industry experience.

A. Research Strategy: Case study

To gain a detailed understanding of Incident Management in a real-world setting, this research utilizes a Case Study strategy. Oates (2006) defines a case study as a strategy that focuses on one instance of the “thing” to be investigated to obtain a rich, detailed insight into its complex relationships and processes.

In this study, the case is defined not as a specific organization but as the expert practitioner’s perspective within the global IT service industry. Rather than analyzing a single company, which limits the scope to one operational model, this study draws upon the cumulative experience of a Program Transformation Director with over 25 years of tenure across multiple major global technology firms. This approach allows the research to analyze Incident Management as a broad industry phenomenon, capturing a longitudinal view of how theoretical frameworks like ITIL intersect with the practical challenges of high-volume, global outsourcing environments

B. Literature Review

This study employs two primary data generation methods: a literature review and a semi-structured expert interview.

A literature review was conducted to establish the theoretical framework and identify existing academic knowledge regarding ITIL and Incident Management. According to Oates (2006), the purpose of a literature review is to show that the researcher is aware of existing work in the chosen topic area and to place the researcher’s work in the context of what has already been published

The review was conducted using the academic database Dalarna University Library (Summon). The search strategy employed keywords such as “Incident Management,” “SWOT,” “SWOT Analysis,” “ITIL 4 alignment,” “ITSM challenges,” “Global IT Operations,” and “Service Level Agreements.” The selection focused on peer-reviewed articles.

C. Expert Interview

The primary empirical data was collected through a structured interview with a Program Transformation Director. The expert brings over 25 years of experience in global IT environments, having worked at reputable companies like Hewlett-Packard (HP) and Wipro. This background ensures the insights capture the practical, realworld context regarding IM challenges, best practices, and the implementation of ITIL 4 on a global scale. The interview questions were designed to directly probe the efficiency, compliance, and resilience of IM processes, covering topics from prioritization and escalation to continuous improvement.

D. Data Analysis

The empirical data collected from the expert interview was analyzed using a SWOT Analysis framework. This method was chosen to provide a structured evaluation of the Incident Management practice by categorizing findings into internal and external dimensions.

- **Internal Factors (Strengths and Weaknesses):** These refer to attributes internal to the organization that are within its control, such as processes, resources, and capabilities (Ghazinoory et al., 2011).
- **External Factors (Opportunities and Threats):** These refer to attributes in the external environment that are outside the organization's direct control, such as market trends, technology shifts, and economic pressures (Helms & Nixon, 2010).

To populate the SWOT framework, the interview transcript was analyzed using a qualitative Thematic Analysis approach as outlined by Oates (2006). This process involved identifying, analyzing, and reporting patterns (themes) within the data. The themes were derived through the following steps:

- **Familiarization:** The analysis began with a thorough reading of the interview transcript to gain a holistic understanding of the expert's perspective.
- **Coding:** Relevant segments of the text were identified and assigned initial codes. For example, statements regarding "meeting SLAs" were coded as *Performance Metrics*, while comments on "staff leaving" were coded as *Attrition*.
- **Categorization:** These codes were then mapped against the four domains of the SWOT Matrix. Positive internal codes (e.g., consistent processes) were categorized as Strengths. Negative internal codes (e.g., premature ticket closure) were categorized as Weaknesses. Positive external codes (e.g., AI adoption) were categorized as Opportunities. Negative external codes (e.g., pricing pressure) were categorized as Threats.

This structured thematic approach ensured that the findings were directly grounded in the participants' realworld experience while remaining aligned with the study's theoretical objectives.

E. Limitations

Several limitations to this study must be acknowledged

- **Single Perspective:** The findings are based on a single expert interview. While the participant has significant experience, the data represents a subjective perspective and may not reflect the views of all IT service providers.
- **Methodological Scope:** As a qualitative case study, the results are specific to the context of large global providers and may not be generalizable to smaller organizations or different IT sectors.
- **Subjectivity:** The categorization of data into SWOT themes relies on the researcher's interpretation of the interview transcript.

V. RESULTS

This section presents the empirical findings derived from the semi-structured interview, categorized using the SWOT framework. The full transcript is available in Appendix A. The findings are analyzed alongside theoretical principles from ITIL 4 and ITSM literature.

Internal Factors	External Factors
Strengths • SLA Compliance: Consistent performance metrics. • Consistent processes: "Fosun" network. • Alignment: Encourage performance.	Opportunities • 1.5 Model: High contact Resolution. • AI Integration: Connecting fragmented systems. • Empowerment: Service deflecting tickets.
Weaknesses	Threats

• Metric Misalignment mature closure tickets. • Communicati Gaps: its unhappy (green SLAs. • Tooling: Implementatic issues with IT tools.	• Commercial Pressure: onflicts in nargin sectors. • Attrition: High pressure causing loss of skilled staff. • Lead Time: Del staffing new con
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Conversely, Threats are largely commercial. In high-volume sectors, clients "challenging... on the price" can undermine the ability to fulfill SLAs. This external pressure exacerbates Staff Attrition, where "support staff is under too much pressure leading to people leaving," threatening the stability of the support model. This highlights a gap in the theoretical framework: while ITIL 4 provides detailed process guidance, it offers limited solutions for the commercial and human capital constraints inherent in global outsourcing (Marrone & Kolbe, 2011).

VI. DISCUSSION

This section interprets the findings presented in the Results, answering the study's research questions by linking the empirical SWOT data to the theoretical framework established by AXELOS (2019), Agutter (2020), and Marrone & Kolbe (2011).

A. Internal Factors: Metrics vs. Quality

The analysis reveals a tension between process consistency (Strength) and metric validity (Weakness). The respondent emphasized that SLA Compliance is the primary driver of performance, focusing on quantitative metrics like "response time" and "speed of resolution." Consistency is maintained globally through "follow the sun (24x7) support," ensuring standardized handovers between regions. This practical finding aligns with Agutter (2020), who notes that standardization is essential for value chain delivery in ITIL 4.

However, a significant divergence from theory was identified regarding service quality. The expert noted that agents sometimes "close tickets... to meet personal targets" even if the issue remains unresolved. This behavior directly contradicts the core ITIL 4 principle of "Focus on Value" (AXELOS, 2019), which states that no action should be taken unless it creates value for the consumer. The expert confirmed that this metric misalignment leads to communication gaps, where "SLAs are met, but the client is not happy".

B. External Factors: Evolution and Risk

The primary Opportunity identified is the evolution toward an L1.5 Support Model. The expert advocated for placing "more knowledge at the team which has 1st contact," thereby improving the "First Contact Resolution rate". This finding strongly supports the ITIL principle of "Optimize and Automate" (Agutter, 2020), as it reduces waste (escalations) and streamlines the incident lifecycle. The integration of AI to "connect multiple systems" further supports this optimization by reducing human toil.

A. RQ1: Key Factors Characterizing Incident Management

The first research question sought to identify the internal and external factors characterizing effective Incident Management. The SWOT analysis reveals that operational effectiveness in large organizations is defined by a tension between process consistency and metric validity.

Internally, the findings highlight that while global standardization (Strength) allows for seamless 24/7 operations, it creates a significant side effect regarding service quality. The expert's observation of premature ticket closure to meet personal targets validates the "Watermelon Effect" described by Marrone & Kolbe (2011), where service metrics appear "green" (compliant) while the actual user experience is "red" (unsatisfied). This suggests that in high-volume environments, internal metrics often incentivize speed over quality, acting as a hidden weakness within the system.

Externally, the practice is characterized by a race between technological evolution and commercial pressure. The emergence of the L1.5 Support Model and AI integration (Opportunities) suggests that the industry is moving away from rigid tiered hierarchies toward more fluid, knowledgebased resolution. However, this is counteracted by commercial pricing pressures (Threats), which the expert identified as a primary cause of staff attrition. This implies that effective Incident Management is not just a technical challenge, but a commercial one: without stable pricing to support skilled staff, process consistency degrades.

B. RQ2: Alignment with ITIL 4 Principles

The second research question analyzed the alignment between practical execution and ITIL 4 principles. The study identifies a distinct divergence regarding the principle of "Focus on Value."

According to Agutter (2020), every operational action must map directly to value creation for the consumer. However, the empirical data shows that operational reality often prioritizes contractual adherence (SLA compliance) over actual value. The "communication gaps" identified in the Weaknesses section demonstrates that when agents prioritize speed (to meet the contract) over communication (to satisfy the user), the ITIL principle is violated.

Conversely, strong alignment is observed regarding the principle of "Optimize and Automate." The expert's advocacy for the L1.5 model and AI-driven knowledge systems mirrors the ITIL 4 recommendation to reduce waste and streamline the value chain (AXELOS, 2019). This suggests that while organizations are successful at implementing the technical aspects of ITIL (tools, automation), they struggle with the cultural aspects (valuecentricity) due to external commercial pressures

Management. Through a SWOT analysis of expert perspectives, the study concludes the following:

RQ1: Factors Characterizing Practice Effective Incident Management is characterized internally by robust global consistency but is undermined by metric-driven behaviors that distort service quality. Externally, it is driven by technological opportunities (AI/L1.5 models) but threatened by commercial pricing pressures that fuel staff attrition.

RQ2: Alignment with ITIL 4 Practical application aligns with ITIL 4 regarding optimization, evidenced by the shift toward AI and flatter support hierarchies. However, it diverges significantly regarding value creation, where rigid SLA contracts often force agents to prioritize speed over user satisfaction.

Recommendations to close these gaps, organizations should:

- Realign Metrics: Shift L1 performance targets from "speed of resolution" to First Contact Resolution (FCR) to align agent incentives with the ITIL goal of value creation.
- Invest in L1.5: Formalize the L1.5 support layer to reduce escalations and utilize AI to empower frontline agents.
- Address Attrition: Mitigate the threat of knowledge loss by incorporating retention strategies into commercial contracts.

VII. CONCLUSION

The objective of this study was to bridge the gap between ITSM theory and the operational reality of global Incident

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IX. APPENDIX A: INTERVIEW TRANSCRIPT

Interviewee Role: Program Transformation Director Experience: 25+ years in Global IT Service Management Format:
Semistructured Interview

Context: Incident Management Terminology Note: The respondent established the following context regarding the operational structure before addressing the SWOT questions.

Respondent: Incident management is organized in levels. L1 is the level where end users connect to; in some cases, that's also called L0. The key point is that support is organized in levels, and every level has different responsibilities. L1 is 1st level support where quick resolution needs to be done within a short amount of time; otherwise, the ticket is assigned to L2 (2nd level support) with more in-depth knowledge, then L3, etc...

Incidents are documented and maintained in incident tickets in an appropriate ITSM/system management tool. Priority is given to incidents, ranging from P4 (low impact, individual) to P1 (major impact, company/building).

a) 1. STRENGTHS

Q1: What processes or practices do you think make incident management particularly effective in large organizations?

Respondent: There are two sides to this. First, the client organization requires responses as agreed in SLAs. In the context of client satisfaction (one of the key drivers of SLA performance), the focus is on response time (when is the end user getting a response) and speed of resolution (when is the incident resolved).

Second, the IT services organization focuses on consistent processes. This takes into account that—depending on support windows (8x5, 10x7)—teams are working in shifts, sometimes in different locations to enable "follow the sun" (24x7) support. Consistency in updating the incident management system and maintaining an up-to-date knowledge base are crucial for effective handling.

Q2: How do established routines or team structures contribute to strong performance during incident handling?

Respondent: They lead to more predictable outcomes and a reduced "learning curve" as incident status is maintained in the system. Controls are implemented to measure how long an incident is in a certain phase to ensure SLA targets are met. Additionally, monthly SLA reporting and meetings are key drivers for regular client satisfaction.

At a personal level, support teams are measured against meeting SLA targets as part of their performance appraisals. Another mechanism that encourages strong performance is individual or team awards, which are published in internal communications.

b) 2. WEAKNESSES

Q3: What are some common challenges or weak points you've observed in incident management processes across organizations?

Respondent: It can happen that at an individual level, an incident ticket is closed while the actual incident has not been resolved. This happens more at the L1 level than above, often because the support agent is trying to meet their personal targets.

Furthermore, sometimes SLAs are met, but the client is not happy. This is mostly linked to communication gaps within the client organization; end users might not be aware of what has been agreed upon regarding response times for lower-priority (P4) incidents.

Q4: Where do you think incident management frameworks like ITIL tend to fall short in real-world application?

Respondent: It is maybe not the ITIL framework which falls short, but the implementation and usage of the related ITSM software package. ITIL should be used as a generic framework and guidance, but the shortfall usually lies in how the tools are applied.

c) 3. OPPORTUNITIES

Q5: What opportunities do you see for improving incident management through new technologies or process innovations?
Respondent: One improvement is providing end users access to "known problems" or a bulletin board on the intranet, fed by the incident management system.

Another key development depends on the cost model: adoption of the L1.5 support model. Unlike the traditional L1 to L2 escalation path, L1.5 places more knowledge at the first point of contact. This results in a higher First Contact Resolution (FCR) rate. AI is also taking a larger role by connecting multiple systems, giving agents access to more sources, which directly supports this L1.5 approach and higher FCR.

Q6: How can organizations strengthen collaboration or knowledge sharing to improve incident handling outcomes?

Respondent: As mentioned, providing end-user access to "known problems" is effective. Quality management reviews, often a separate function at the account level, help improve incident handling at regular intervals. Additionally, providing appropriate AI training to support agents is an opportunity. Finally, having appropriate Service Manager counterparts at the client's organization is key for collaboration.

d) 4. THREATS

Q7: What external or internal risks could threaten the effectiveness of an organization's incident management process?

Respondent: For some clients, contracts and SLAs are challenging because the client challenges the IT services company on price. This often happens in high-volume, low-margin sectors like retail, which impacts the ability to fulfill the SLA.

Another risk is staffing. For new contracts, there must be sufficient lead time to get the right staffing in place. If there isn't sufficient time, it creates a risk for existing clients if staff are moved to prioritize the new contract. While SLA waivers can mitigate penalties initially, high pressure can result in higher attrition (staff leaving the company). Consistency and stability in the delivery organization are key; if rewards and career growth are not organized well, trained support agents will leave.

Q8: How can organizations prepare for or mitigate risks linked to increasing system complexity or cyber threats?

Respondent: For cyber threats, most IT service organizations have a separate part of their delivery organization focusing completely on this in Security Operation Centers (SOCs).

For increasing system complexity, there is no "one size fits all" answer. It starts with having the right knowledge within the right teams. Therefore, internal delivery assurance during the contract preparation phase is key; different capabilities within the delivery organization must take responsibility to ensure they can support the specific contract complexity.

At a personal level, support teams are measured against meeting SLA targets as part of their performance appraisals. Another mechanism that encourages strong performance is individual or team awards, which are published in internal communications.